GE VERNOVA

GridOS Alarm

0.7.0

User Guide

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Change History

Date	Change Description
2025-May-14	GridOS Alarm 0.2.0: • This is a new document for GridOS Alarm 0.2.0.
2025-Jun-18	GridOS Alarm 0.3.0: • Added Areas of Responsibility.
2025-Aug-14	GridOS Alarm 0.4.0: • Updated Alarms and Events Page with details about acknowledging an alarm. • Added Saving Table Views.

2025-Oct-27	GridOS Alarm 0.5.0:
	• Added:
	  ○ Alarm Management
	  ○ Alarm Processing, includes content from the *GridOS Alarm Installation Guide*, and new content for overlays and deletion.
	  ○ Time Format.
	• Updated Areas of Responsibility to move some technical content to the *GridOS Alarm Installation Guide*.
	• Editorial and formatting updates.

2025-Dec-16	GridOS Alarm 0.6.0: • Content moved between Alarm/Event Table and Customizing the Data Table. • Updated Customizing the Data Table to move some technical content to the *GridOS Alarm Installation Guide*. • Updated Alarm Processing with details on multiple acknowledgement and multiple delete. • Added Sympathetic Auto-Acknowledgment. • Editorial and formatting updates.

2026-Mar-23	GridOS Alarm 0.7.0:
	• Added:
	  ○ Alarm List Size Management.
	  ○ User Login and Logout.
	• Editorial and formatting updates.

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1. Alarms and Events Page

This chapter provides an overview of the **Alarms and Events** page.

1.1. Access the Alarms and Events Page

To access the Alarm and Events page:

1. Click **GridOS Alarm** from the applications dropdown list if running in the App Shell, or open `https://<scada-domain>/deui-alarm/` in a web browser.

   > **Note**
   >
   > If you are not logged in, opening the Alarm UI URL redirects you to the login page and prompts you to enter your GridOS user credentials. Enter your username and password, then click **Login** to access the Alarm UI.

2. Click either the **Alarms** or **Events** tab based on the details you want to view.

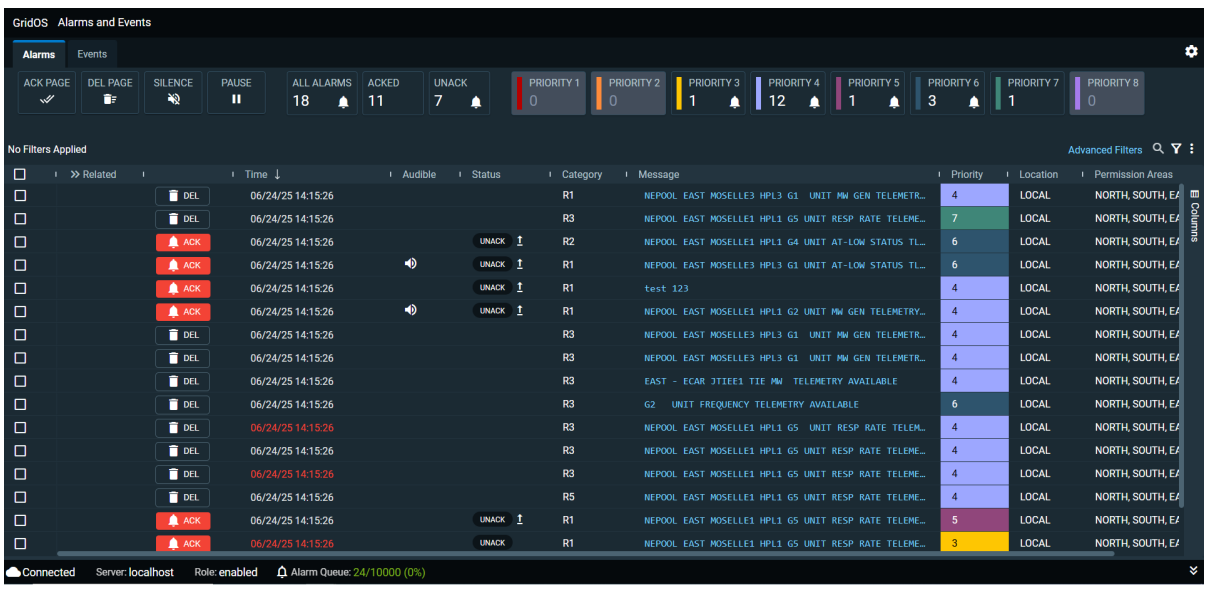


Figure 1. Alarms and Events Page

The **Alarms and Events** page includes the following main areas:

- **Toolbar:** Includes action buttons and predefined filters.

  For more information, see Toolbar.

- **Data table:** Each row represents an alarm or event (depending on the selected tab). You can customize the data table view based on your needs.

  For more information, see Customizing the Data Table.

- **Status bar:** Shows details about your connection and available data.

  For more information, see Status Bar.

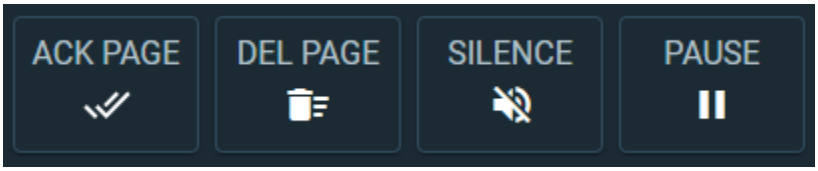
> **Tip**
>
> To log out, click the user icon at the top right of the page and click **Logout** from the dropdown menu.

The following sections describe each of these areas.

1.2. Toolbar

1.2.1. Action Buttons

The toolbar contains the following action buttons (from left to right):



- **Acknowledge Page:** Acknowledges all unacknowledged alarms that are visible on the current page.

  > **Note**
  >
  > If one or more alarms are selected, the button changes to **Acknowledge Selected**.

- **Delete Page:** Deletes all acknowledged alarms on the current page.

  > **Note**
  >
  > If one or more alarms are selected, the button changes to **Delete Selected**.

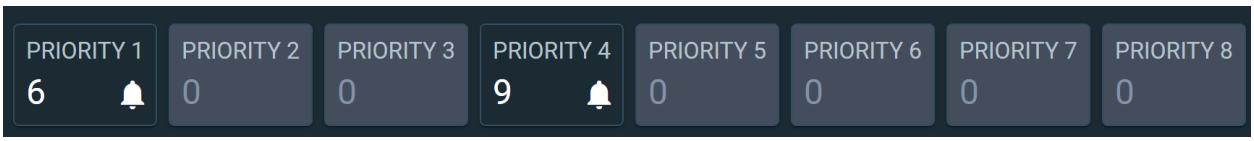
- **Silence:** N/A. Functionality is not yet available.
- **Pause:** Pauses alarm data refresh. A watermark is displayed on the Alarm UI, indicating that data is paused.

Acknowledging an alarm is an asynchronous operation, which usually completes quickly. However, there may be times during high activity, a failover, or over poor network connections where it may take several seconds for the operation to complete. If a failure occurs, a notification displays any relevant error messages.

1.2.2. Predefined Filters

1.2.2.1. Priority

Use the predefined filters at the top of the page to view alarms by priority. By default, filters are off, showing all priority values.

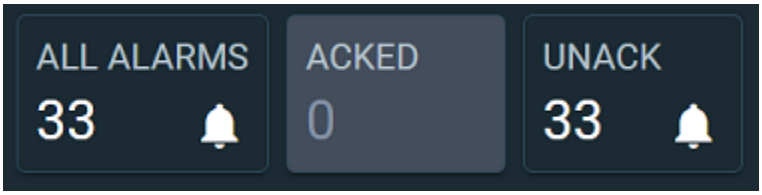


Select one or more priority buttons to filter the table and show only alarms with the selected priorities.

An alarm bell icon appears next to the alarm count if there are unacknowledged alarms for a given priority. The count shows the total number of alarms for the priority.

1.2.2.2. Acknowledgement

Use the predefined filters at the top of the page to view all alarms (removing all filters), or to show only acknowledged or unacknowledged alarms.



The following options are available:

- **All Alarms:** Shows all alarms by clearing any active filters.

  If there are unacknowledged alarms, an alarm bell icon appears next to the alarm count.
- **Acked:** Shows only acknowledged alarms.
- **Unack:** Shows only unacknowledged alarms.

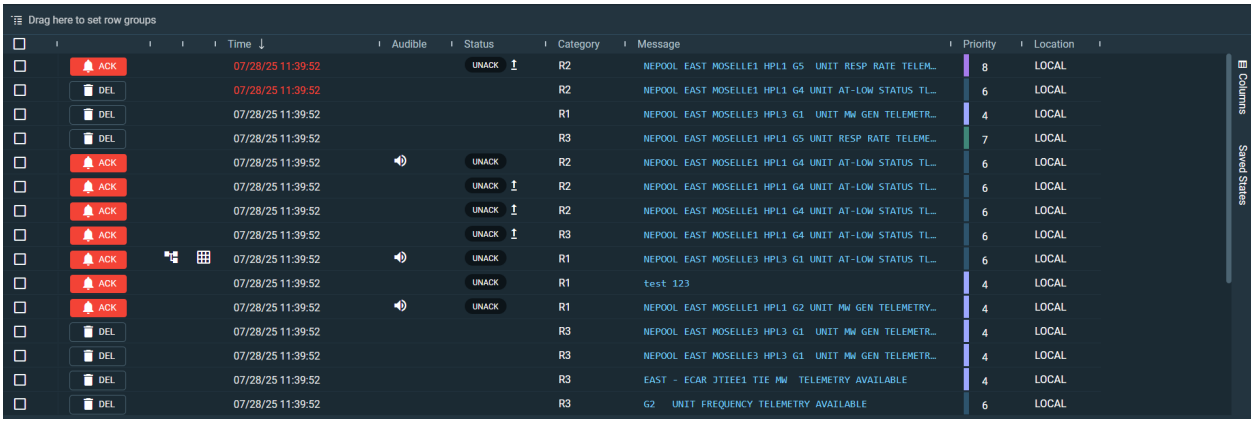
1.3. Alarm/Event Table

Alarm and event details are shown in a table. You can modify the view based on your needs.

For more information, see Customizing the Data Table.

> **Note**
>
> The alarms and events that are visible depend on various factors.

> For more information, see Alarms Displayed in the UI.



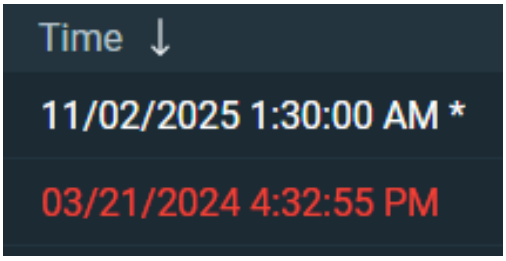
1.3.1. Time Format

The Alarm UI displays date and time in a localized format based on your preferences.

For more information about how to set these preferences, refer to the "Alarm UI Setup" chapter in the *GridOS Alarm Installation Guide*.

1.3.2. DST Indicator

If an alarm is created during the repeated hour caused by Daylight Saving Time (DST), an indicator (an asterisk by default) appears next to the time value in the **Time** column.



1.4. Status Bar

This section describes the alarm status bar at the bottom of the page.

1.4.1. Connection Status

The connection status indicator shows the current connection status by monitoring the alarm data service heartbeat time.

If GridOS Alarm is not connected, the status bar will appear as below:

1.4.2. Server

The **Server** field indicates which data server GridOS Alarm is currently retrieving data from.

1.4.3. Alarm Queue

The **Alarm Queue** field shows the current number of alarms, the alarm queue size, and the percentage used.

If the percentage used is above the configured amount for caution or warning, it is color-coded yellow or red accordingly.

For example:

- alarmQueueCautionPercent: 75
- alarmQueueWarningPercent: 90

Given the above configuration, the alarm queue status indicator turns yellow (caution) when the alarm queue exceeds 75%, and turns red (warning) when it exceeds 90%.

> **Note**
>
> The Alarm Queue information always reflects the status of the Alarm database (real-time or standby).

1.5. UI Settings

Click the gear icon at the top-right of the UI to access the **Settings** panel.

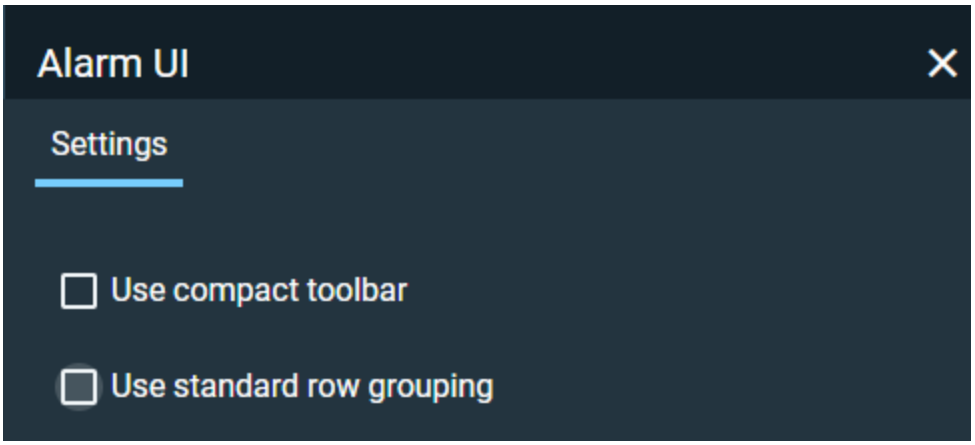
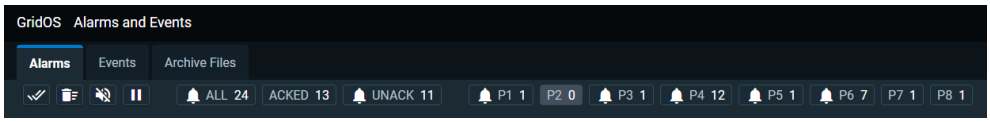


Table 1. Settings Panel Options

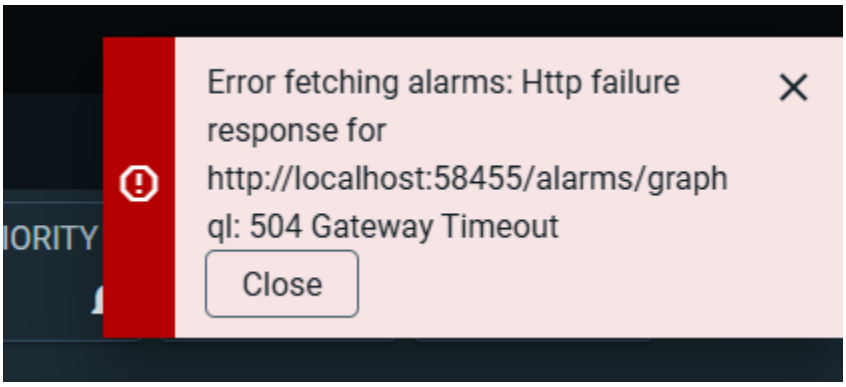
Setting	Purpose
Use compact toolbar	Switch the toolbar to its compact version. 
Use standard row grouping	Change the table to use standard row grouping, which allows grouping rows by dragging columns into the grouping drop area. For more information, see Grouping Related Alarms.

1.6. Additional Functionality

1.6.1. Notification Messages

Notification messages are shown in the upper-right portion of the Alarm UI. The types of messages shown are controlled by the `notifyOnMessageTypes` configuration variable in the Alarm UI application configuration file.

By default, only error messages are displayed.



Click **Close** to dismiss the notification message popup.

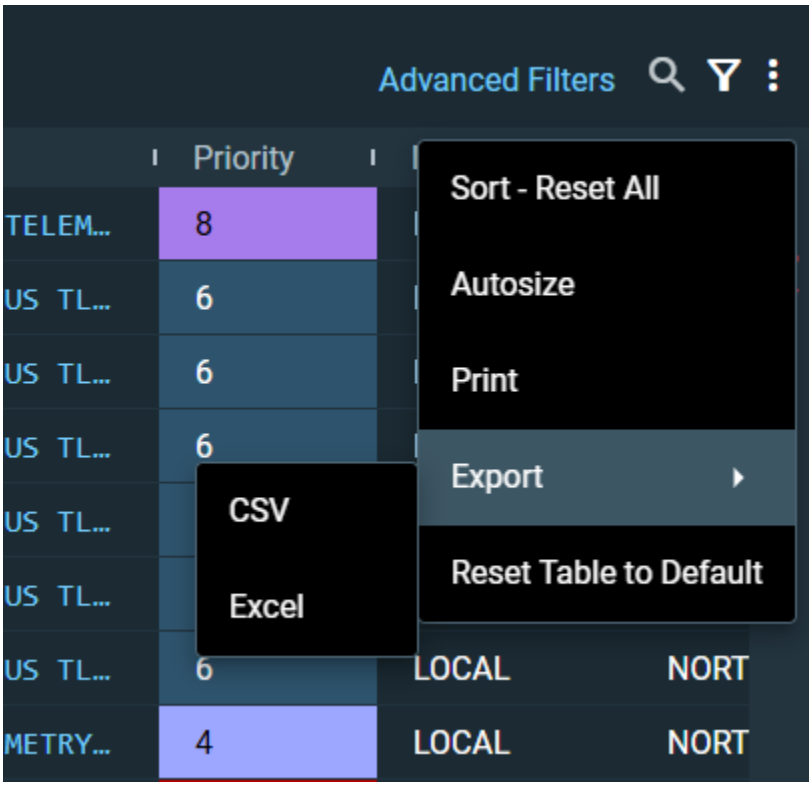
The color on the message box indicates the severity of the message:

- **Red:** Error
- **Yellow:** Warning
- **Blue:** Information
- **Green:** Success

1.6.2. Excel and CSV Export

Click the three-dots menu at the top-right of the table and click **Export** to download table data in either Excel or CSV format.

When exporting data, only rows currently included in the grid are exported.



1.6.3. User Login and Logout

If you are not logged in, opening the Alarm UI URL redirects you to the login page and prompts you to enter your GridOS user credentials.

Enter your username and password, then click **Login** to access the Alarm UI.

To log out, click the user icon at the top right of the page and select **Logout** from the dropdown

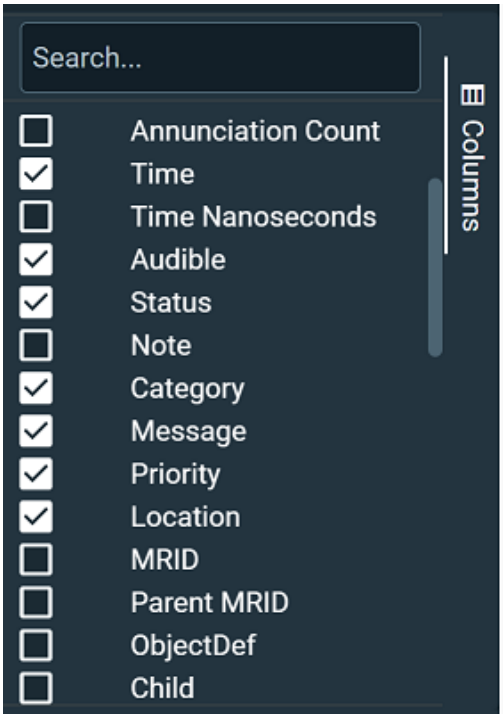
menu.

2. Customizing the Data Table

In addition to the predefined filters available at the top of the **Alarms and Events** page, various options are available to customize the view of the data table so that you can view and analyze the information based on your needs.

2.1. Visible Columns

You can use the **Columns** panel on the right side of the table to show or hide columns.



> **Note**
>
> Columns can be configured so that they cannot be hidden by you. In such cases, the checkbox is disabled in the **Columns** panel.

2.1.1. Available Columns

The following table lists the available columns in the Alarm UI, along with a brief description.

Column Name	Description	Shown By Default?
Selection Column	A column with checkboxes to select rows, including a header checkbox to select all rows.	✓

Column Name	Description	Shown By Default?
Related Column	When grouping alarms by parent/child relationship, this column displays the number of alarms and unacknowledged alarms (in parentheses) in the group.	✓
Acknowledge/Delete Button Column	Contains a button for acknowledging or deleting the alarm.	✓
Open SLD	An icon column to open the Single Line Diagram (SLD) for the associated object.	✓
Open Tabular	An icon column to open a tabular view for the associated object.	✓
Annunciation Time	Displays the time of annunciation in a formatted date-time string.	✗
Annunciation Time Nano	Stores the nanoseconds part of the annunciation time.	✗
Annunciation Count	Displays the count of annunciations.	✗
Time	Displays the time of the alarm in a formatted date-time string. Includes conditional styling for abnormal alarms. By default, the time text is displayed in red for abnormal alarms.	✓
Time Nanoseconds	Stores the nanoseconds part of the time field.	✗
Audible	An icon column indicating whether the alarm is audible. A volume icon appears when `true`.	✓
Status	Displays the status of the alarm.	✓
Note	An icon column for viewing or editing notes attached to the alarm. Visibility is conditional based on the presence of a note.	✓
Category	Displays the category of the alarm.	✓
Message	Displays the alarm message.	✓
Priority	Displays the priority of the alarm.	✓
Location	Displays the location of the alarm.	✓
MRID	Stores the unique identifier of the alarm.	✗
Parent MRID	Stores the unique identifier of the parent alarm.	✗
ObjectDef	Stores the object definition type (for example, POINT and ANALOG).	✗
Child	Indicates whether the alarm is a child alarm.	✗
Time Acknowledged	Displays the time when the alarm was acknowledged in a formatted date-time string.	✗
Color	Stores the color associated with the alarm.	✗
Category Color	Stores the background color of the category.	✗
Category Foreground Color	Stores the foreground color of the category.	✗
Composite Key	Stores the composite key for the alarm.	✗
Composite ID	Stores the composite ID for the alarm.	✗
Domain	Stores the domain of the alarm.	✗
Field Time	Displays the field time of the alarm in a formatted date-time string.	✗

Column Name	Description	Shown By Default?
Field Time Nanoseconds	Stores the nanoseconds part of the field time.	✗
Logs	Stores logs associated with the alarm.	✗
Permission Areas	Displays the permission areas associated with the alarm as a comma-separated string.	✓
Unack Areas	Displays the unacknowledged areas associated with the alarm as a comma-separated string.	✗
Unpurged Areas	Displays the unpurged areas associated with the alarm as a comma-separated string.	✗
Relative Time	Displays the relative time of the alarm using a custom renderer.	✗
Sequence Number	Stores the sequence number of the alarm.	✗
Source	Stores the source of the alarm.	✗
Station	Stores the station ID associated with the alarm.	✗
User	Stores the user ID associated with the alarm.	✗
Abnormal	Indicates whether the alarm is abnormal.	✗
Alarm MRID	Stores the unique identifier for the alarm instance.	✗
Client State	Stores the client state of the alarm.	✗
Exception Definition	Stores the exception definition for the alarm.	✗
Non-deletable	Indicates whether the alarm is non-deletable.	✗
Object MRID	Stores the unique identifier for the object associated with the alarm.	✗
PC Name	Stores the PC name from which users perform an action on an alarm, such as acknowledging or deleting it.	✗
Severity	Stores the severity level of the alarm.	✗
URL	Stores the URL associated with the alarm. This is used for opening the operator guide associated with the alarm.	✗

2.2. Filtering

This section describes the filters that are available in GridOS Alarm.

Except for advanced filters, any number of filters can be applied to the table at one time.

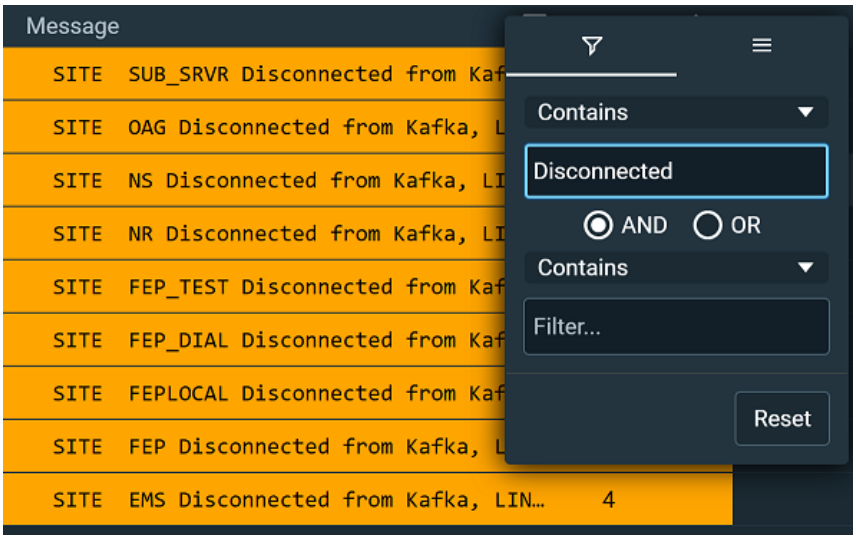
2.2.1. Column Filters

2.2.1.1. Text Filter

Some columns, like the Message column, allow free-form text. To filter such a column, hover the mouse cursor to the right of the column name and select the filter dropdown from the header

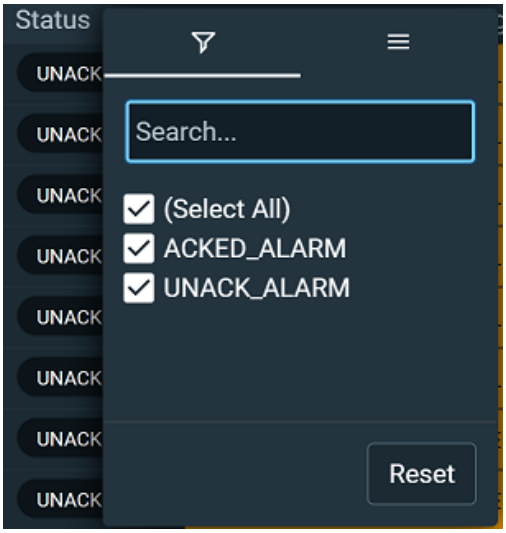
menu.

The column filter menu allows you to enter text and select various options such as Contains, Starts With, and Equals. Once entered, the grid updates to show relevant rows.



2.2.1.2. Set Filter

Some data types in the grid, like alarm status, lend themselves to filtering via a set.

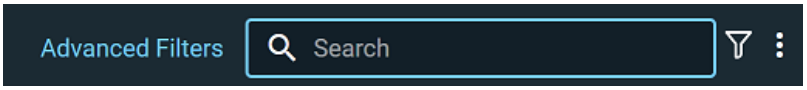


The data shown in the filter list represents all values in that column that are currently visible (or not filtered out). Set filters are applied by clicking the checkboxes next to them.

Typing in the search box does not change the selected filters but helps you find set filters by name.

2.2.2. Table-Level Text Filter (Search)

The Table search box, accessible from the search icon located at the top-right of the table, lets you filter the entire table by any text.

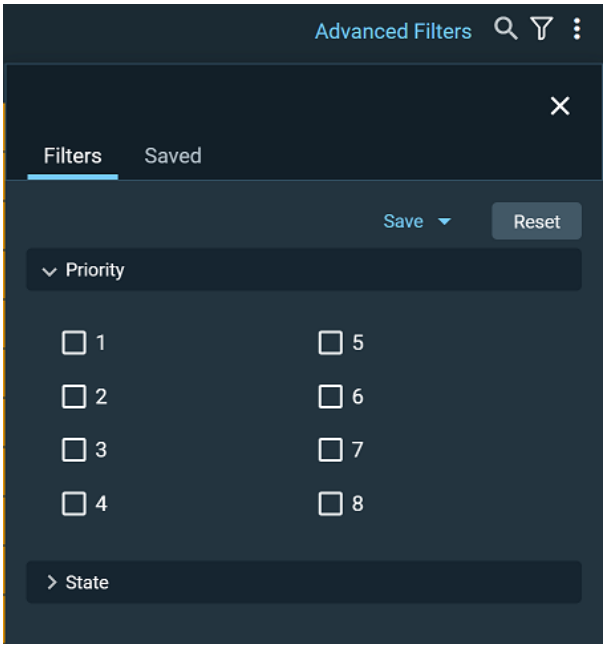


The text can be leading, trailing, or appear anywhere in the filtered fields. The filter applies to any text-based data like description and category.

> **Note**
>
> Certain fields, like formatted time fields, might be stored internally as milliseconds but only appear as a localized string for presentation. These columns (Time, abnormal, etc.) are not considered by the filter.

2.2.3. Filter Panel

The Filter panel allows you to save, delete, and edit filters.



To create a filter:

1. Open the **Filter** panel using the Filter icon at the top-right of the table.
2. Select the filter options you want from the options presented in the filter tab.

To save the filter:

1. Select the **Save** menu option. After saving, the filter will appear in the **Saved** tab.

> **Note**
>
> Filter options shown in the Filter tab are configured in the table configuration file in the section named `sideFilters`.

2.2.4. Advanced Filtering

Advanced Filtering allows for complex filter conditions to be applied across columns. When an advanced filter is enabled, an input field will appear where you can enter a filter expression.

To create an advanced filter:

1. Select **Advanced Filters** at the top-right of the table.
2. Type the name of the column for which you want to add a filter. As you type, the list of available columns will appear. After you have chosen the column, press Enter.
3. Select from the list of filter operators and press Enter.
4. Enter the value you want to filter by and press Enter. The table will be filtered by the expression.

   Complex filter expressions can be created using `AND` and `OR` conditions.

> **Note**
>
> The advanced filter and regular column filters cannot be active at the same time. Enabling the advanced filter disables column filters.

2.3. Sorting

2.3.1. Column Sorting

Most columns shown in the UI can be sorted. Clicking the column header will cycle through three sort modes: Ascending, Descending, and None.

> **Note**
>
> Some columns are not sortable, such as the action icon columns for acknowledging and deleting alarms.

2.3.2. Multi-Column Sorting

Multiple columns can be sorted at the same time by pressing the Shift key while selecting additional sort columns. A numeric indicator shows the column sort order.

Time 1 ↓	Field Time 2 ↓
04/29/25 12:12:33	01/09/24 23:18:34
04/28/25 21:52:42	01/09/24 23:18:34
04/28/25 15:56:03	04/28/25 15:56:03
04/28/25 15:56:03	04/28/25 15:56:03

As seen in the example above, the **Time** column is sorted first and the **Field Time** column is sorted second.

2.3.3. Default Sort Order

The default sort order for the alarm table is determined by the column configuration in the table configuration file. Each column has a `sort` field which can be set to `desc` or `asc`. The order that the sort columns appears in the table configuration file determines the order.

For the default alarms table, the sort order is as follows:

- Time
- Time Nanoseconds

> **Note**
>
> Columns can participate in sorting even when they are hidden.

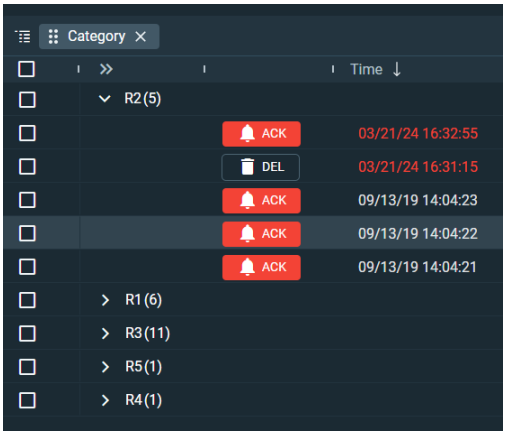
The default sort order can be restored at any time by resetting the table to the default state using the **Sort-Reset All** option in the table menu.

2.4. Standard Row Grouping

By default, the table data is grouped according to the parent/child relationship of alarms. If an alarm has children, the parent and child alarms are shown in a hierarchical view.

The table can also be grouped using standard row grouping, which allows you to group alarms by any column. This is useful for organizing alarms based on specific criteria, such as category or status.

The image below shows an example of standard row grouping, where alarms are grouped by the **Category** column.



2.4.1. Selection and Operation Logic

When related alarms are grouped, the behavior of actions (for example, acknowledging an alarm) and selections for a parent alarm vs. a child alarm is configurable.

If the `parentOpsApplyToChildren` variable is set to `true` in the application configuration file, then operating on a parent alarm attempts to apply that same operation to any child alarm with a matching state. For example, assume an alarm is unacknowledged and has five children. Of those five, one is also unacknowledged, and four are acknowledged. Acknowledging the parent will also acknowledge the single child.

Parent alarms can be expanded, and child alarms can be individually selected for operations.

If you want all child alarms to be selected when a parent alarm is selected, set the `groupSelectsChildren` variable to `true` in the corresponding table configuration file.

2.4.2. Constraints

Because alarms may have many children, or there may be many alarms selected at a point when an Acknowledge or Delete operation is made, it is necessary to constrain the number of alarms that can be acknowledged or deleted at a given time. For example, attempting to acknowledge 1,000 alarms in a single operation will overwhelm the alarm subsystem.

Additionally, it may be desirable to simply limit how many alarms can be operated at a given time.

For example, setting a small value such as "5" would prevent someone from acknowledging a thousand alarms by mistake.

For these reasons, a configuration parameter exists in the alarm, event, and history configurations. This value is called `maxAlarmsPerTransaction` and can be set to any desired value. The default value is 50.

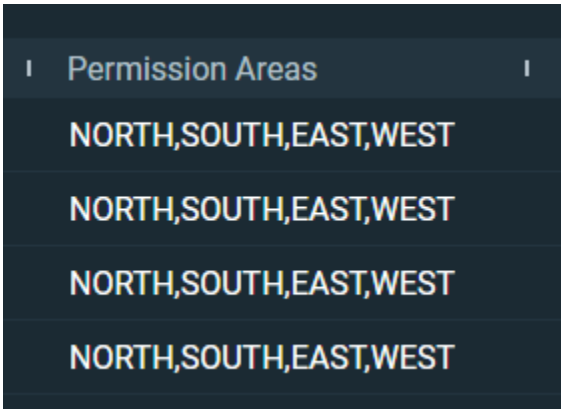
> **Important**
>
> Setting this to a large value can overwhelm the alarm subsystem and may result in losing

incoming alarms.

2.5. Areas of Responsibility

Areas of Responsibility determine which Alarms and actions are visible to an operator. The operator can further filter alarms based on area in order to focus on specific areas of the grid.

The alarms in the table can be filtered by clicking on the **Permission Areas** column header and selecting the areas you want to filter by.



Tip

For more information about AORs, refer to:

- Permissions in the *GridOS Alarm Installation Guide*.
- Partial Acknowledgement Due to AORs

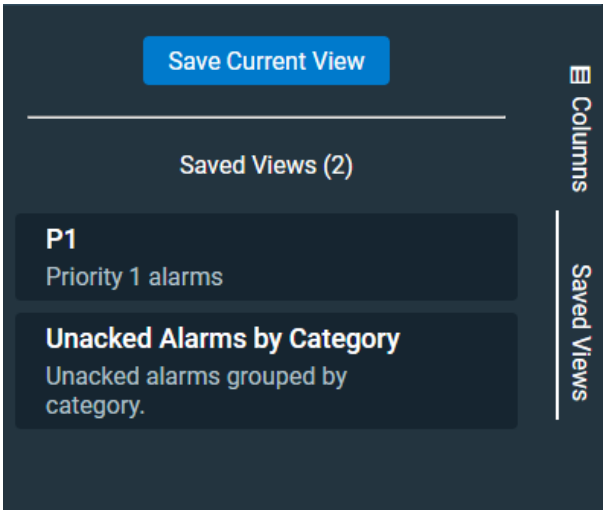
2.6. Settings Persistence

The Alarm UI allows you to customize the data table view, including visible columns, filters, sorting, and grouping. These settings are saved per user in the user's preferences, and are restored when the Alarm UI is loaded.

2.6.1. Saving Table Views

You can save the current view of the data table, including filters, sorting, column visibility, and grouping, as a named view using the Table View Panel. This lets you quickly restore the table to a specific configuration.

Note: Advanced Filtering and saving/restoring table views cannot be used together. The Table View Panel is disabled when Advanced Filtering is active.



Accessing the Table View Panel:

- Click the **Saved Views** tab at the right of the table.

To save a table view:

1. Adjust the table settings (filters, sorting, etc.) as needed.
2. Click **Save Current View** in the Table View Panel.
3. Provide a name and optional description, then click **Save**.

Saved table views are stored in the user's preferences and are restored when the Alarm UI is loaded, maintaining configurations across sessions.

To restore a saved view:

- Click the desired view in the Table View Panel.

To delete a saved view:

- Hover over the view and click the **Delete** button.

To reset to the default view:

- Use the **Reset Table to Default** option in the table menu. This clears all applied changes to the table, and resets the table to its default configuration.

3. Alarm Management

GridOS Alarm provides a consolidated view of alarms and events triggered in the GridOS system across applications. You may then review the alarms and determine the appropriate next course of action. Acknowledgment is the action taken by an operator to indicate that an event has been noted. Deletion removes the alarm from the list to make room for new alarms. The alarm remains in the system for historical purposes.

3.1. Alarms Displayed in the UI

The first step in managing alarms is understanding which alarms are displayed in the Alarm user interface.

This is determined by the following factors:

- **User Permissions**: Only the alarms which the logged-in user has valid permission for (both action-related and AOR-related) are displayed on the **Alarms and Events** page.

  For more information, refer to the "Permission" chapter in the *GridOS Alarm Installation Guide*.

- **Alarm Processing**: Events are processed before being included in the **Alarm and Event** lists, which impacts how they are displayed on the **Alarms and Events** page.

  For more information, see Alarm Processing.

- **Page Filters or Settings**: The **Alarms and Events** page has multiple filters and settings to adjust the view of alarms.

  For more information, see Alarms and Events Page and Customizing the Data Grid.

3.2. Acknowledging Alarms

To acknowledge an alarm:

- Click the **Ack Page** or **Ack Selected** button in the toolbar.

  > **Note**
  >
  > If one or more alarms are selected, the **Ack Page** button changes to **Ack Selected**.

  -OR-

- Click the **ACK**

button in the Alarm table.

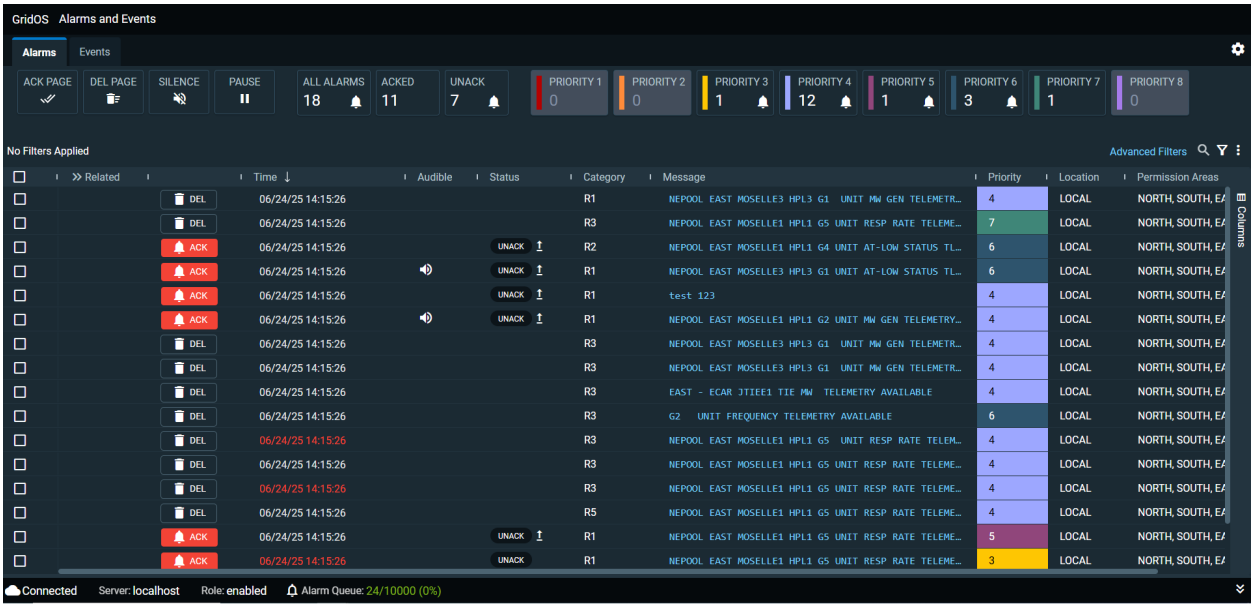


Figure 2. Alarms and Events Page

3.2.1. After an Alarm is Acknowledged

When an alarm is acknowledged:

- The system checks the Exception table configuration for the alarm type to determine if any additional actions are required.
- If auto-deletion is enabled for the alarm type, the alarm is deleted after being acknowledged (but only for the areas assigned to the user. See the next section for more details).
- If multiple acknowledge is enabled for an alarm type, acknowledging a single alarm will also acknowledge all related alarms.

For more information about alarm acknowledgement processing, see Alarm Processing.

3.2.2. Partial Acknowledgement Due to AORs

Acknowledgements are applied only to the AORs which the user performing the action has permission for. If an alarm was associated with other AORs, unacknowledged areas remain, and the alarm is considered only partially acknowledged.

Partially acknowledged alarms appear in the UI as acknowledged or unacknowledged based on the permissions of the logged-in user:

- If the alarm is acknowledged for all the areas that the user has permissions for, the alarm appears as fully acknowledged to that user.
- If the alarm is unacknowledged for at least one area that the user has permissions for, the alarm appears as unacknowledged.

- If multiple acknowledge is enabled for the alarm type, all related alarms are acknowledged for the areas assigned to the user.

For example, suppose Alarm X is associated with AORs [1 , 2]. User A has permission for AOR 1, and User B has permission for AOR 2. If User A acknowledges Alarm X, only the portion of the alarm related to AOR 1 is acknowledged. Since the AOR 2 portion remains unacknowledged, Alarm X will still appear as unacknowledged to User B. However, because all AORs assigned to User A are acknowledged, the alarm will appear as acknowledged to User A in the UI. The alarm is considered partially acknowledged until all associated AORs have been acknowledged by users with the appropriate permissions.

3.3. Deleting Alarms

To delete an alarm:

- Click the **Del Page** or **Del Selected** button in the Toolbar.

> **Note**
>
> If one or more alarms are selected, the **Del Page** button changes to **Del Selected**.

-OR-

- Click the **DEL** button in the Alarm Table.

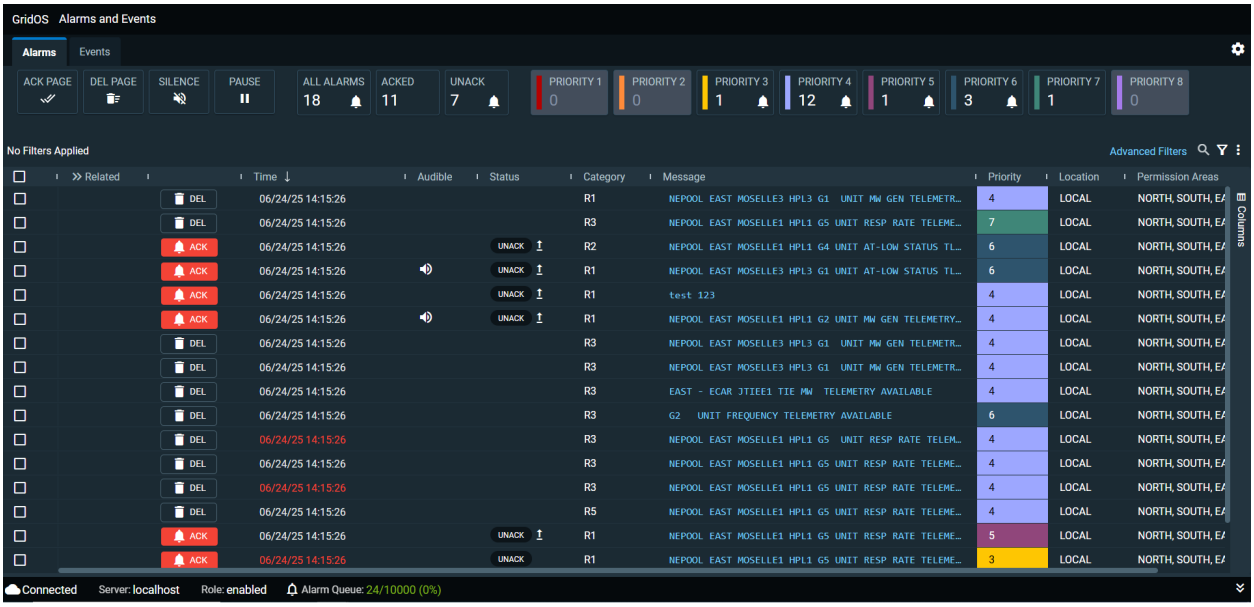


Figure 3. Alarms and Events Page

When a delete request is received, the service checks the following before proceeding with the delete operation:

- Alarm is not abnormal, or `abnormalDeletion` is not set on the category.
- `nonDeletable` flag is not set for the alarm.
- Whether multiple delete is enabled for the alarm type; if it is, deleting a single alarm will also delete all related alarms.

For more information about alarm deletion processing, see Alarm Processing.

3.3.1. Partial Delete Due to AORs

Deletes are applied only to the AORs which the user performing the action has permission for. If an alarm was associated with other AORs, unpurged areas remain, and the alarm is considered only partially deleted.

Partially deleted alarms appear in the UI or not based on the permissions of the logged-in user:

- If the alarm is deleted for all the areas that the user has permissions for, the alarm does not appear in the list for that user.
- If the alarm is not deleted for at least one area that the user has permissions for, the alarm appears in the list.

3.4. Alarm List Size Management

The alarm list is automatically managed to prevent it from growing beyond a configurable maximum size. When the number of alarms reaches a configured threshold, the system automatically purges the oldest and lowest priority alarms to keep the list within bounds.

3.4.1. How Alarms Are Selected for Purging

When the alarm list reaches the configured threshold, the system selects alarms for removal in the following priority order:

1. **Acknowledged alarms first** — Alarms that have already been acknowledged are selected before unacknowledged alarms, as these are the safest to remove. Within this group, alarms are ordered by:
    - Lowest priority first.
    - Oldest timestamp first (within the same priority).
2. **Unacknowledged alarms as supplement** — If there are not enough acknowledged alarms to meet the purge quota, the remaining slots are filled with unacknowledged alarms using the same ordering (lowest priority, oldest first). These alarms are force-acknowledged by the system prior to removal.

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> **Note**
>
> Alarms flagged as non-deletable (`nonDeletable`) are never selected for automatic purging.

3.4.2. How Purged Alarms Are Processed

Each alarm selected for purging is processed through the same pipeline as a normal user-initiated action. The system acts under the actor name **SYSTEM** and applies full acknowledgement and deletion processing, including any exception-level behaviours such as cascaded acknowledgements or auto-deletion on acknowledgement.

The processing differs depending on the alarm's state at the time of purging:

- **Already-acknowledged alarms** — A delete action is issued and processed through the full pipeline. A single event log entry is written recording that the alarm was deleted by SYSTEM.
- **Unacknowledged alarms** — An acknowledge action is issued first, followed by a delete action. Each action is processed through the full pipeline and each produces its own event log entry. All normal side effects of acknowledgement apply (for example, cascaded multiple-acknowledgement, auto-delete on acknowledgement).

As a result, for a large purge, operators may see a significant number of new entries in the event log — up to two entries per unacknowledged alarm (one for the system acknowledgement and one for the deletion), and one entry per already-acknowledged alarm. This is by design and provides a full auditable trail of every alarm the system removed.

3.4.3. System Event on Purge

In addition to the individual event log entries described above, a single summary event is written to the event list each time an automatic purge occurs. The summary event indicates:

- The total number of alarms removed from the alarm list.
- The number of those alarms that were unacknowledged and were force-acknowledged by the system prior to removal.

> **Note**
>
> The summary event and the individual action events are written to the event list, not the alarm list.

For more information about alarm sizing configuration, refer to the "Alarm Service Parameters" chapter in the *GridOS Alarm Installation Guide*.

4. Alarm Processing

The Alarm Process determines which incoming alarms need to be updated in the realtime Alarm List. The Alarm List grows and shrinks based on when alarms occur and return to normal. The Alarm List is dynamic and filtered and sorted for each operator based on the AORs assigned to the operator.

All Alarms and Events (including Audit, AlarmAck, AlarmDel) will be added to the realtime Event List. The Event List is a fixed size log where the oldest items are overwritten when new entries are received. Actions done on alarms (such as acknowledgements and deletes) create events so that there is an audit trail of the alarm.

Advanced processing features operate on Related Alarms, defined as those whose alarm exception types are under the same exception group (object definitions) in the alarm model.

> **Tip**
>
> For more information about the various processing configurations discussed on this page, refer to the "Alarm Model" chapter in the *GridOS Alarm Installation Guide*.

> **Tip**
>
> You can configure the default values for Areas of Responsibility (AORs), Category, and Exception Definition.
>
> For more information, refer to the "Alarm Service" chapter in the *GridOS Alarm Installation Guide*

4.1. Areas of Responsibility (AOR) Validation

The Alarm Service validates AORs provided on all new alarms before adding them to the database.

- When an alarm is generated, the service checks each provided AOR against the modeled AORs (retrieved from the AOR Manager).
- Any invalid or unknown AORs are stripped from the alarm data.
- If no valid AOR is assigned, the service applies a default AOR.

4.2. Auto-Acknowledgements

Some incoming alarms are automatically acknowledged by the system based on their configuration. When an alarm event is processed, the service checks if the alarm should be automatically acknowledged before making it to the alarm or event list.

The processing logic for auto-acknowledgements first examines the `useApiOptions` from the Exception table configuration.

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- If `useApiOptions` is `true`, the system checks the `gridCoreKind` field on the incoming event, and uses that property as well as the `autoAcknowledgement` flag from the Exception table to determine state.
- If `useApiOptions` is `false`, the system only uses flags from the Exception table to determine the state.

The following diagram illustrates the processing logic for auto-acknowledgements.

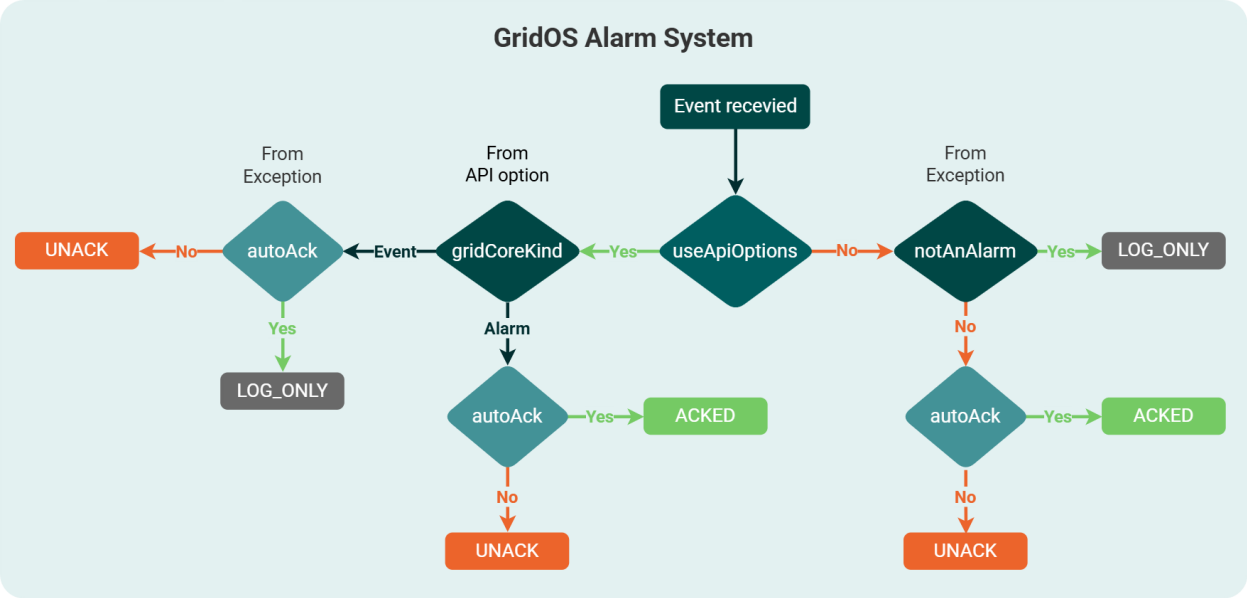

Figure 4. Auto-Acknowledgement Processing Logic

4.2.1. Sympathetic Auto-Acknowledgment

If both `autoAcknowledgement` and `sympatheticAutoAcknowledgement` are enabled, the system automatically acknowledges any AORs whose most recent related alarm has already been acknowledged. Related alarms are matched sequentially.

The system searches for the next most recent alarm that occurred before the incoming alarm. This ensures that acknowledgments are applied only to relevant areas, preventing over-acknowledgment and maintaining precise alarm state control. If no related alarms are found, all AORs remain unacknowledged.

For example, if Alarm X occurs with AORs [A, B], and an operator acknowledges only AOR A, then when the next related Alarm Y arrives with AORs [A, C], the system automatically acknowledges AOR A (because it was acknowledged in Alarm X), while AOR C remains unacknowledged (because it was not part of the previous alarm).

4.3. Overlay Acknowledgement

When `overlayAcknowledgement` is enabled for an exception group, any existing related alarms are

acknowledged when a newer alarm is received.

4.4. Multiple Acknowledgement

When the `multipleAcknowledgement` flag is enabled on the exception group associated with an alarm type, acknowledging a single alarm also acknowledges all related alarms. The system identifies related alarms and processes the acknowledgement for each one.

4.5. Deletion

Some alarms are automatically deleted by the system based on their configuration. When an alarm event is processed, the service checks if the alarm should be automatically deleted before making it to the alarm, in which case it will appear only in the event list.

Incoming alarms are automatically deleted if they are acknowledged after the auto-acknowledgement processing, described above and `autoDelete` is enabled for the alarm type in the Exception table configuration.

When `overlayDelete` is enabled for an exception group, any existing related alarms are acknowledged and deleted when a newer alarm is received.

4.6. Multiple Delete

When the `multipleDelete` flag is enabled on the exception group associated with an alarm type, deleting a single alarm also deletes all related alarms. The system identifies related alarms and processes the delete for each one.